Uncertainty-Aware GPU Hour Allocation via Conformal Risk Control

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1. Abstract

In shared GPU clusters, jobs must be assigned a time allocation before their true runtime is known. If a job exceeds its allocation, it is **killed** and **all prior compute is wasted**. Current practice uses fixed multiplicative buffers such as 2× predicted runtime, applying uniform margins regardless of per-job uncertainty.

We combine Conformalized Quantile Regression (CQR) with Conformal Risk Control (CRC) to produce per-job allocations with formal finite-sample guarantees on expected overrun.

On the Alibaba PAI trace (732K jobs), we achieve **up to 50% cost savings over fixed buffers** at matched coverage, with **26.3 μs** allocation time per job.

2. Problem

- **Scheduling:** which job runs next, on which GPUs.
- **Allocation:** how many GPU-seconds to reserve.

Asymmetric cost model:

$$\mathrm{cost}(a, y) = \max(a - y, 0) + R \cdot \max(y - a, 0)$$

Waste from over-allocation incurs linear cost, while under-allocation kills the job and loses all prior compute.

Why fixed buffers fail

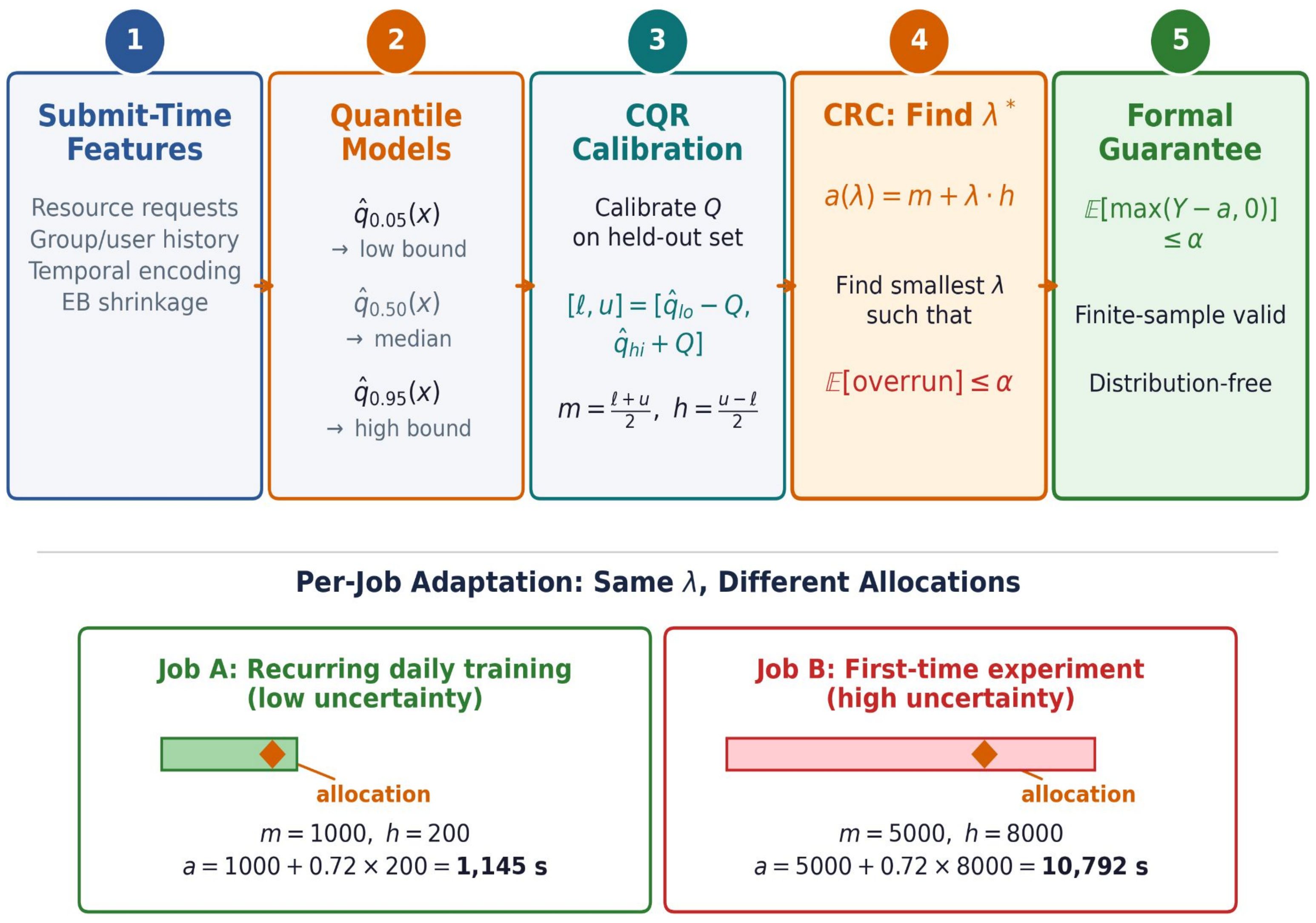
Buffer	Coverage	Cost	Alloc. (s)
200%	87.9%	23,704	9,679
1000%	95.8%	42,002	35,557
5000%	99.0%	163,684	164,854
20000%	99.8%	645,283	649,792

Even a **20,000% buffer does not reach 100% coverage**; 224 jobs have actual/predicted ratio above **342×.**

Dataset & Setup

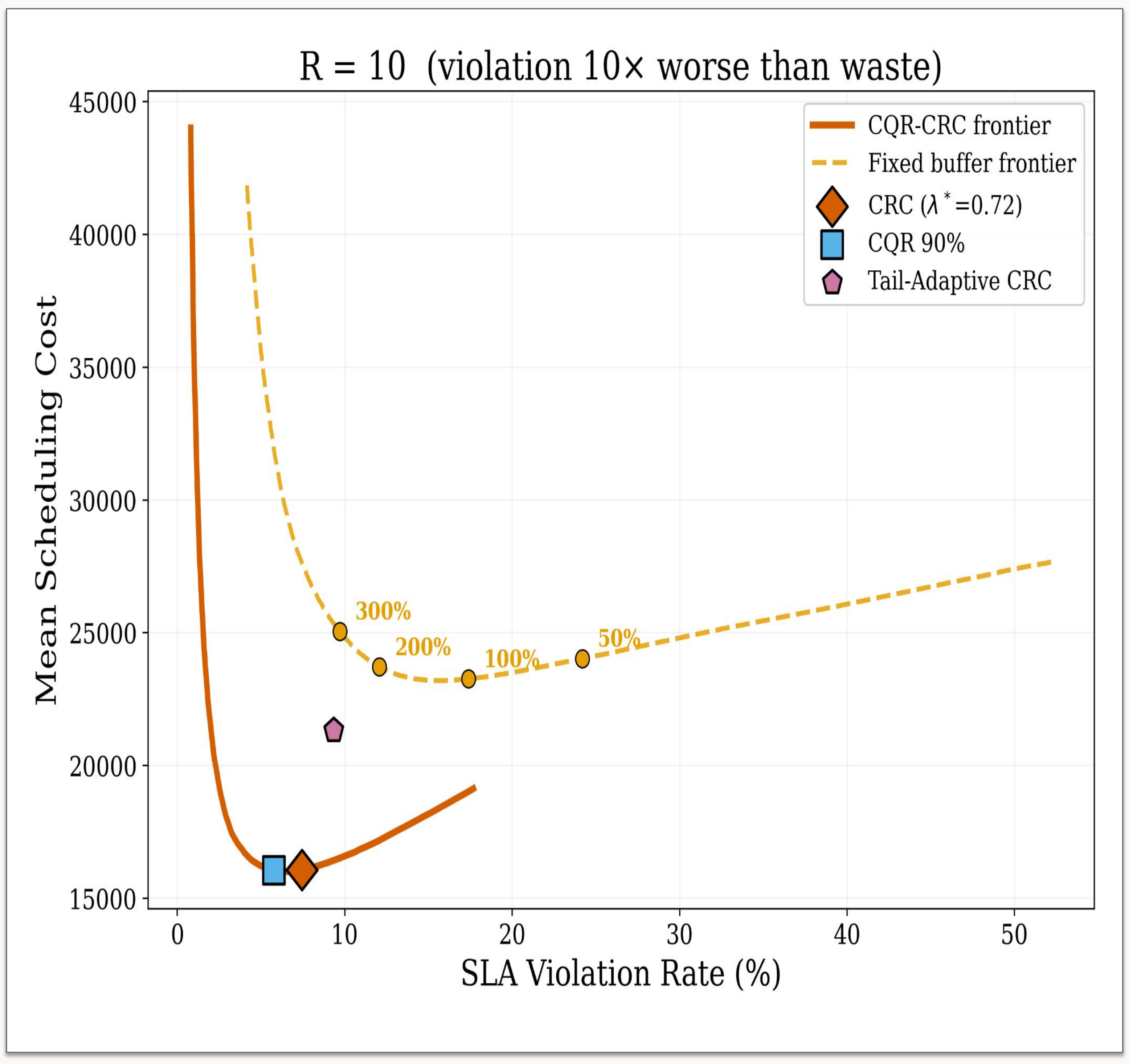
- Alibaba PAI-2020 trace via ATLAS benchmark (ICLR 2026).
- 732,355 jobs, large-scale production trace.
- 31 submit-time features; no post-execution leakage.
- Chronological split: 55% / 15% / 15% / 15% (train / val / test / held-out).
- LightGBM quantile regressors (α = 0.05, 0.50, 0.95).
- Cross-validation on Microsoft Philly cluster trace.
- Per-job allocation time: 26.3 μs.

3. Method



- **CQR** produces per-job prediction intervals, narrow for predictable jobs and wide for uncertain jobs.
- **CRC** chooses the **smallest λ** such that expected overrun is bounded by α.
- Allocation formula: $a = m + \lambda \cdot h$, where m is the interval midpoint and h is the interval half-width.
- Key insight: **overrun decreases monotonically** in λ, enabling CRC guarantees; because total cost is not monotone, we decompose the problem.
- **Tail-adaptive variant**: apply risk control by runtime bucket (<10 min, 10–60 min, 1–6 hr, >6 hr) to protect expensive long-running jobs.

4. Pareto Dominance



Our approach Pareto-dominates fixed buffers: at every operating point, it achieves lower mean scheduling cost for the same SLA violation rate.

Scheduling Impact: Max-Stretch

Method	ρmax at R = 50
EDF-P baseline	2,461.5
Buffer 100%	2,225.2
Tail-Adaptive ours	2,145.6

- Tail-adaptive allocation gives the lowest worst-case job stretch.
- Protecting expensive long jobs prevents queue-blocking “kills” in EDF.
- These improvements come from using uncertainty to reserve extra time only where it matters most.

Results

Cost Savings at Matched Coverage

R	Coverage	Ours	Best Buf.	Savings
2	82.4%	5,823	6,724	**13.4%**
5	87.5%	10,399	14,021	**25.8%**
10	92.5%	16,056	26,249	**38.8%**
20	96.2%	24,490	49,027	**50.0%**
50	**98.3%**	**43,432**	**—**	**N/A***

* No fixed buffer achieves 98.3% coverage.

Kill Rates by Runtime

Runtime	Point	Buf 100%	Ours	n
<10 min	35.4%	4.3%	**0.6%**	52,991
10–60 min	60.9%	22.1%	**7.1%**	27,851
1–6 hr	72.6%	33.4%	**14.7%**	22,489
6–24 hr	79.1%	42.9%	**34.3%**	5,707
>24 hr	**98.3%**	**86.2%**	**74.9%**	**816**

- **Top 1% of jobs account for 46.8% of total scheduling cost.**
- The largest gains come from protecting long, uncertain jobs rather than uniformly buffering all jobs.
- Uncertainty-aware allocation reduces both kill rates and wasted GPU hours.

Future Work

- **Real-World Experiment: Tiresias** scheduler integration for cluster-wide metrics.
- **Uncertainty-aware scheduling** using interval width in priority decisions.
- **Per-class risk** parameter R by job type.
- **Extensions to multi-resource scheduling** and heterogeneous GPU pools.

References

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